Practical 1 - Gold exploration (data modified after Arndt 2017)

Background

You are working as a geologist for a small mining outfit specialized in gold. Your latest prospect (area with suspected enrichment of a commodity) is located in the remote jungle of Indonesia. Previous regional mapping by remote sensing and ground truthing has identified gold mineralization along veins and intrusions.

You are in charge of a team tasked with narrowing down the location of gold deposits in the area. Access is extremely difficult, mostly along streams and rivers, and locally away from these. Consequently, a previous stage has collected as many stream samples as possible and measured gold concentration in the sediment. The resulting map is attached and gave you an indication which streams intersect the source deposit. An area of veins running NW-SE (highlighted orange) was identified.

In a second stage, after narrowing down the search area, you took rock and soil samples and measured their gold concentration. Can you identify the most likely location of the original deposit using a contour map of gold concentrations?

Although strongly simplified for the exercise, this would be a typical task in exploration of unknown areas, usually requiring several weeks of field and laboratory work.

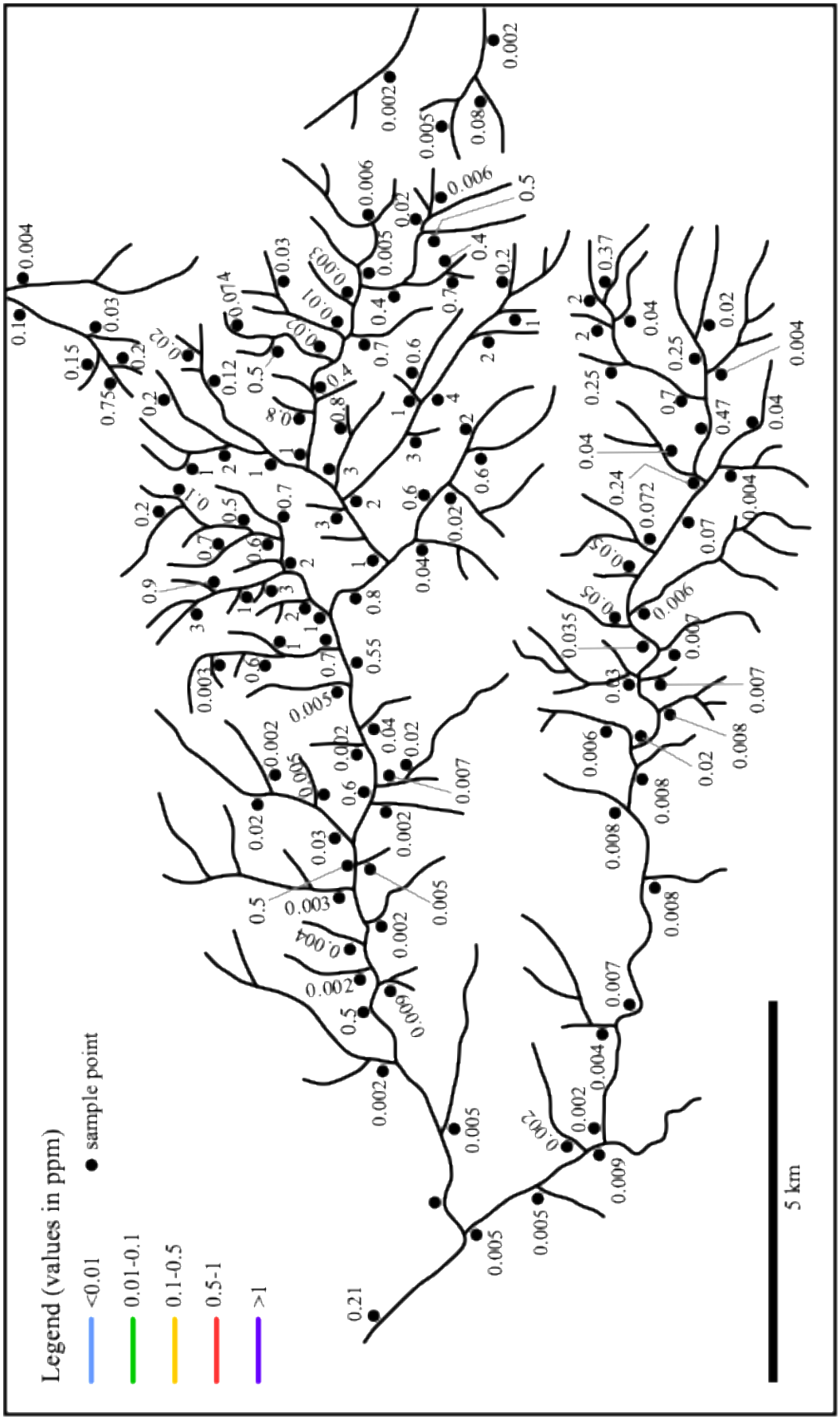
Exercise

You are being given two maps.

Map 1 shows the local stream network in the prospect area, together with the gold concentration measured in stream samples (in ppm). This was already by color-coding stream segments and has identified a prospective area, which you are now investigating in more detail.

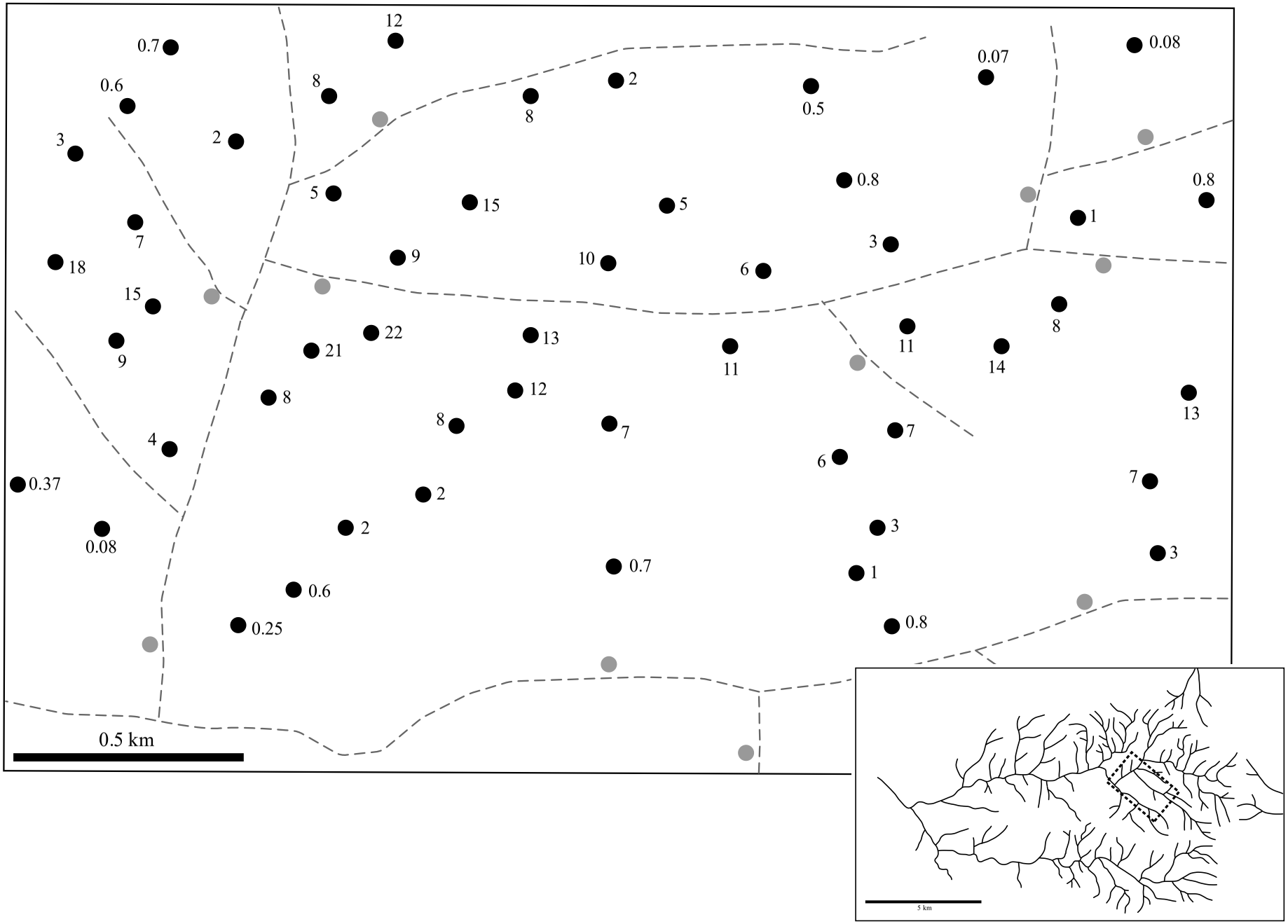
Map 2 is a blow-up of a small area (indicated by dotted rectangle in map 1) where rock and soil samples were taken. Again, gold concentrations are given in ppm. For reference, the stream network and location of stream samples are provided in grey.

Make a contour map from the soil gold concentrations. Use 0.1, 1, 5, 10, 20 ppm contour lines. Where would you drill?



Map 1

Map 2


0.7
12
0.6
8
8
2
0.5
0.07
0.08
3
2
5
15
5
0.8
0.8
7
1
18
9
10
6
3
15
8
22
13
11
11
14
21
9
12
8
8
7
7
13
4
6
0.37
2
7
2
3
0.08
3
0.7
1
0.6
0.25
0.8
0.5 km
5 km